Task 2: CPU Scheduling

Q2 Evaluation & Round Robin Implementation Report

Student Name: Shiva Sai Krishna S | **Register Number:** 1911260318

Course: CSA0421 - Operating System | **Bloom's Level:** K5 - Evaluate (8 Marks)

GitHub Repository: https://github.com/krishshiva721/CSA0421-OPERATING-SYSTEM/tree/main/CSA04--Assignment2

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1. Problem Context & Challenge Summary

> **Scenario:** Cloud Video-Encoding Service handling short jobs (thumbnails, 2s) and long jobs (full films, 20m). Small jobs wait behind long ones under basic scheduling. Goal: evaluate algorithms and implement Round Robin.

2. Complete C Code: `round_robin.c`

```c
#include <stdio.h>
#define N 4

int main(void) {
    const char *name[N] = {"P1", "P2", "P3", "P4"};
    int arrival[N] = {0, 1, 2, 3};
    int burst[N]   = {8, 4, 9, 5};
    int quantum = 3;
    int queue[64], head = 0, tail = 0, added[N] = {0};
    int left[N], finish[N];
    for (int i = 0; i < N; i++) left[i] = burst[i];
    int time = 0, done = 0;

    queue[tail++] = 0; added[0] = 1;
    printf("Gantt:");
    while (done < N) {
        int p = queue[head++];
        int run_time = (left[p] < quantum) ? left[p] : quantum;
        printf(" %s(%d-%d)", name[p], time, time + run_time);
        time += run_time;
        left[p] -= run_time;

        for (int i = 0; i < N; i++) {
            if (!added[i] && arrival[i] <= time) {
                queue[tail++] = i; added[i] = 1;
            }
        }
        if (left[p] > 0) {
            queue[tail++] = p;
        } else {
            finish[p] = time; done++;
        }
    }
    double tw = 0, tt = 0;
    printf("\nProcess  TAT  WT\n");
    for (int i = 0; i < N; i++) {
        int tat = finish[i] - arrival[i];
        int wt = tat - burst[i];
        tt += tat; tw += wt;
        printf("%-7s  %2d  %2d\n", name[i], tat, wt);
    }
    printf("Average TAT %.2f, average WT %.2f\n", tt / N, tw / N);
    return 0;
}
```

3. Execution Output

```
Gantt: P1(0-3) P2(3-6) P3(6-9) P4(9-12) P1(12-15) P2(15-16) P3(16-19) P4(19-21) P1(21-23) P3(23-26)
Process  TAT  WT
P1        23  15
P2        15  11
P3        24  15
P4        18  13
Average TAT 20.00, average WT 13.50
```

4. Time Quantum Study & Evaluation

Quantum (q)	Avg WT	Avg TAT	Context Switches	Evaluation & Impact
q = 1	14.25	20.75	25	High context switch overhead; degrades actual CPU execution efficiency.
q = 3	13.50	20.00	10	Optimal balance between fast response time and lower switching overhead.
q = 5	12.75	19.25	6	Good throughput, but short job latency starts increasing slightly.
q = 20	12.25	18.75	4	Degenerates into FCFS; short jobs get trapped behind large video encodings.

5. Architectural Recommendation

> **Recommendation: Multilevel Feedback Queue (MLFQ)**
> A Multilevel Feedback Queue with 3 tiers (Queue 0: q=2s RR, Queue 1: q=8s RR, Queue 2: FCFS) is recommended. New interactive/short video thumbnail jobs start at Queue 0 and complete in 2 seconds without waiting behind 20-minute encodings. Long jobs automatically sink to Queue 2.

6. Alternative Concept & Complexity Analysis

Metric / Parameter	Single Round Robin Queue	Multilevel Feedback Queue (MLFQ)
Scheduling Complexity	`O(1)` queue head pop	`O(1)` per queue priority level scan
Short Job Waiting Time	High if quantum is large or queue is long	Near zero (immediate execution in top queue)
Starvation Risk	None (Pure Round Robin)	Possible for low queue without ageing; eliminated via period boost